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Adaptive Two-Stage Designs for Single-Arm Phase IIA Cancer Clinical Trials

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SUMMARY. The main purpose of a phase IIA trial of a new anticancer therapy is to determine whether the therapy has sufficient promise against a specific type of tumor to warrant its further development. The therapy will be rejected for further investigation if the true response rate is less than some uninteresting level and the test of hypothesis is powered at a specific target response rate. Two-stage designs are commonly used for this situation. However, in many situations investigators often express concern about uncertainty in targeting the alternative hypothesis to study power at the planning stage. In this article, motivated by a real example, we propose a strategy for adaptive two-stage designs that will use the information at the first stage of the study to either reject the therapy or continue testing with either an optimistic or a skeptic target response rate, while the type I error rate is controlled. We also introduce new optimal criteria to reduce the expected total sample size.

KEY WORDS: Phase IIA cancer trials; Sample size; Two-stage designs.

1. Introduction

Phase IIA cancer clinical trials are conducted to determine whether a new anticancer therapy has sufficient activity against a specific type of tumor to warrant further (phases IIB and III) development. These trials usually have a single arm, with the primary endpoint being the tumor response rate defined clinically (Leventhal and Wittes, 1988). The true response rate is denoted by p. The null hypothesis is H_0: $p \leq p_0$, where p_0 is chosen to indicate an uninteresting response rate. If the null hypothesis is true, we require the type I error rate to be controlled at a prespecified level. We also require the study to be powered at a specified alternative hypothesis H_A: $p \geq p_*$, where $p_* > p_0$ and is chosen to indicate a response rate that the new therapy may achieve and is worth further investigation. For ethical and logistic considerations, a two-stage design may be used for these phase IIA trials (Herson, 1984).

In practice, p_0 is easily specified, but p_* is not. In the early 1960s, when anticancer agents had low activity, Gehan's two-stage design, in which $p_0 = 0$, was used widely (Gehan, 1961). Later, Simon generalized p_0 and introduced "optimal" and "minimax" designs (Simon, 1989). Other designs which extend Simon's can be found in Green and Dahlberg (1992), Chen (1997), Herndon (1998), and Shuster (2002). All these designs and testing procedures considered some sort of optimal early stopping for futility or efficacy. Bayesian designs have also been proposed recently by, for example, Tan and Machin (2002).

In this article we are concerned with the specification of p_* (and the associated power) and propose a strategy for adaptive two-stage designs. In designing clinical trials, especially in early phase investigation of new treatments, we often face uncertainty in assuming the variability of the response variable and/or the magnitude of the treatment effect, as discussed in, for example, Shih (1992), Gould and Shih (1998), and Zucker et al. (1999). This can be demonstrated by the following example which motivated this article. A study to investigate the weekly administration of vinorelbine, bleomycin, and gemcitabine (VBG) combination therapy was considered at the Cancer Institute of New Jersey for treating patients with recurrent or refractory Hodgkin's disease (HD). Because each of the single agents has been known to be active against HD when given alone, p_0 was set at 0.40. To distinguish the combination from the single agents, the principal investigator targeted power for the response rate of the VBG combination therapy at $p_* = 0.60$ based on his initial expectation. However, seeing the literature on other sorts of combinations, the co-investigators debated and expressed uncertainty with the initial expectation and indicated that a $p_* = 0.50$ to 0.55 response rate would still merit further phase IIB and III investigations. The question is: How can we design the study in some optimal way given the uncertainty in specifying the treatment effect p_*? If we use 0.60, then the study may be underpowered if the true rate is close to 0.50. On the other hand, if we use 0.50 or 0.55, patient resources may be wasted and recruitment will take an unnecessarily long

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time if the true response rate is really greater than or equal to 0.60.

A natural way to resolve this dilemma is to choose p_* with some flexibility. With the proposed adaptive designs, we allow the initial target value to be high and then lower it if necessary, or vice versa, depending on the result of the first stage of the study, and then determine the number of patients needed for the second stage accordingly. This idea is comparable to the "sample size reestimation" or "internal pilot study" situation, where a continuous variable is considered but the variance is uncertain for the study power or sample size (Shih, 2001; Wittes, 2002). For the binary variable case, the variability of response is a function of the treatment-effect size; see discussion in Shih and Zhao (1997) for phase III double-blind comparative studies. Here, in considering phase IIA single-arm cancer studies, we pay more attention to the feasibility of the study in terms of the study duration and sample size, the possibility of early acceptance of the null hypothesis, and the optimality of designs to minimize the exposure of patients to early investigation of new treatment. We will give a full exposition of our adaptive two-stage approach in the following sections.

2. Adaptive Two-Stage Designs

We first describe the proposed designs with the parameters, constraints, and optimality criteria, then prove the existence of solutions. We also discuss an algorithm to find feasible and optimal solutions.

2.1 *Description of Procedure*

Recall that p_0 is the maximum uninteresting response rate. Let p_1 and p_2 be two (low/high) choices of the target response rates, where $p_0 < p_1 < p_2$. In this design, n_1 patients will be recruited to the first stage. The number of patients needed for the second stage will depend on the number of observed responses in the first stage. Let x be the number of the observed responses in the first stage out of n_1 patients. The procedure is as follows:

- If $x \le s_1$, stop the trial with failure to reject H_0.
- If $s_1 < x \le r_1$, power the study at $(1-\beta_1)$ for $p_* = p_1$ and enter $m_2 = m - n_1$ additional patients into the study;
  Reject the treatment (more accurately, reject $p > p_1$) if the total number of responses $\le s$ out of m patients.
- If $x > r_1$, power the study at $(1-\beta_2)$ for $p_* = p_2$ and enter $n_2 = n - n_1$ additional patients;
  Reject the treatment (more accurately, reject $p > p_2$) if the total number of responses $\le r$ out of n patients.

We require that type I error $\le \alpha$. Although not required, it is reasonable to let $\beta_1 \ge \beta_2$ in practice because we would like to have higher power for detecting more improvement of the new therapy (p_2 vs. p_0), and, from the feasibility aspect, we need to compromise the power somewhat for less improvement (p_1 vs. p_0) because the sample size cannot be too big for a phase II study. This capping of sample size (hence compromising the power somewhat) for less effect has also been used in the sample size reestimation methods for feasibility (see, e.g., Gould and Shih, 1998). In the drug development sense, if a treatment is "rejected," the appropriate conclusion is to reject the alternative hypothesis, not to accept the null hypothesis. Similarly, if the treatment is accepted for further study, the appropriate conclusion is to reject the null hypothesis, not to accept the alternative hypothesis. Rejecting H_0 at phase IIA is to allow the drug testing to continue in phase IIB/III trials with more patients and using time to progression or death as the primary endpoint.

2.2 *Optimal Adaptive Two-Stage Designs*

From the above setting, the adaptive two-stage design parameters are s_1, r_1, n_1, s, m, r, and n. If the true response probability is p, then the probability of accepting H_0 is

$$G(s_1,r_1,n_1,s,m,r,n,p) = B(s_1,n_1,p) + \sum_{x=s_1+1}^{\min(r_1,s)} b(x,n_1,p)B(s-x,m_2,p) + \sum_{x=r_1+1}^{\min(r,n_1)} b(x,n_1,p)B(r-x,n_2,p),$$

where $b(\cdot,\cdot,\cdot)$ and $B(\cdot,\cdot,\cdot)$ are the PDF and CDF of the binomial distribution, respectively. First, $G(s_1, r_1, n_1, s, m, r, n, p)$ has to satisfy the following error constraints:

$$G(s_1,r_1,n_1,s,m,r,n,p) \ge 1-\alpha, \quad \text{for } p \le p_0; \tag{1}$$

$$G(s_1,r_1,n_1,s,m,r,n,p) \le \beta_1, \quad \text{for } p \ge p_1; \tag{2}$$

$$G(s_1,r_1,n_1,s,m,r,n,p) \le \beta_2, \quad \text{for } p \ge p_2. \tag{3}$$

Second, because we would like to have smaller probability to dismiss the new therapy when p is larger, it is desirable that the design parameters s_1, r_1, n_1, s, m, r, and n be chosen so that the function $G(s_1, r_1, n_1, s, m, r, n, p)$ is a decreasing function of p. When $G(s_1, r_1, n_1, s, m, r, n, p)$ is a decreasing function of p, the above inequalities are equivalent to the following:

$$G(s_1,r_1,n_1,s,m,r,n,p_0) \ge 1-\alpha; \tag{4}$$

$$G(s_1,r_1,n_1,s,m,r,n,p_1) \le \beta_1; \tag{5}$$

$$G(s_1,r_1,n_1,s,m,r,n,p_2) \le \beta_2. \tag{6}$$

We will call $(s_1, r_1, n_1, s, m, r, n, p)$ a feasible solution if it satisfies the error constraint (1), (2), and (3).

In the next section we show properties of $G(s_1, r_1, n_1, s, m, r, n, p)$. Some of these properties are in turn used to show the existence of a feasible solution $(s_1, r_1, n_1, s, m, r, n)$ to our design, and some of these properties are used to find optimal solutions among the feasible solutions. The optimality is defined in terms of minimizing the sample sizes as follows: When the true response rate is p, the expected sample size is

$$E(N\,|\,p) = n_1 + \{(B(r_1,n_1,p) - B(s_1,n_1,p))m_2 + (1-B(r_1,n_1,p))\}n_2.$$

The following four optimality criteria for s_1, r_1, n_1, s, m, r, and n may be considered:

- Optimality Type 1 (O1): EN_0 is smallest; or

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- Optimality Type 2 (O2): $\max\{EN_i \mid i = 0, 1, 2\}$ is smallest; or
- Optimality Type 3 (O3): $\max(n, m)$ is smallest among all feasible solutions and EN_0 is smallest among such solutions; or
- Optimality Type 4 (O4): $\max(n, m)$ is smallest among all feasible solutions and $\max\{EN_i \mid i = 0, 1, 2\}$ is smallest among such solutions,

where $EN_i = E(N \mid p_i)$ for $i = 0, 1, 2$.

O1 and O3 are extensions to Simon's "optimal design" and "minimax design" criteria, respectively. When $p_1 = p_2$ and $\beta_1 = \beta_2$, then $r_1 = s_1$, $s = r$, $m = n$, and O1 and O3 reduce to Simon's designs. As in Simon (1989), the O1 design achieves reduction in EN_0 by having a smaller sample size at the first stage. This is desirable for ethical considerations in case of inactive therapy. On the other hand, the O3 design has a smaller maximum sample size. The O2 and O4 designs are new; they are natural extensions of O1 and O3, respectively. In O2 and O4 we consider the smallest expected number of patients not only under the null hypothesis but also under the alternatives as well. For O2 and O4, compared to O1 and O3, respectively, a larger sample size is needed at the first stage, but they possess the advantage of more savings in the total sample size. A larger sample size needed at the first stage is reasonable to ensure a sound decision based on the interim data for adaptive designs, and hence may not be a drawback for O2 and O4. More discussion is given when we continue the numerical example.

2.3 *Properties of Function $G(s_1, r_1, n_1, s, m, r, n, p)$*

The following properties of the function G are useful to show the existence of solutions and to find the optimal solutions.

(a) For any fixed s_1, r_1, n_1, p, $q = r/n$ and $q' = s/m$, as $n \to \infty$ and $m \to \infty$,

$$G(s_1, r_1, n_1, s, m, r, n, p) \to \begin{cases} 1 & \text{if } p < \min(q, q'), \\ B(r_1, n_1, p) & \text{if } q < p < q', \\ 1 + B(s_1, n_1, p) - B(r_1, n_1, p) & \text{if } q' < p < q, \\ B(s_1, n_1, p) & \text{if } p > \max(q, q'). \end{cases}$$

(b) $G(s_1, r_1, n_1, s, m, r, n, p)$ is an increasing function of s for any fixed p, s_1, r_1, n_1, m, r, and n.

(c) $G(s_1, r_1, n_1, s, m, r, n, p)$ is an increasing function of r for any fixed p, s_1, r_1, n_1, s, m, and n.

The proofs of these properties can be found in the Appendix.

2.4 *Existence of Solutions*

The existence of $(s_1, r_1, n_1, s, m, r, n)$ that satisfies constraints (1)–(3) for any given $0 < p_0 < p_1 < p_2 < 1$, α, β_1, and β_2 can be shown based on the property (a) in the preceding section and the proof is given in Appendix A2. For feasible solutions we will follow Simon (1989) to search for finite m and n within a range based on what a single-stage design would require. More about the search for feasible and optimal solutions are discussed in the next section.

2.5 *Algorithm to Find Feasible and Optimal Solutions*

The detailed algorithm to find feasible and optimal solutions are available from the authors. Briefly, for specified value of p_0, p_1, p_2, α, β_1, and β_2 we first search for feasible m and n within the range of 0.85–1.5 times of what a single-stage design would require for the sample size if p_0 is tested against p_1 or p_2, respectively. This is based on the similar strategy used by Simon in his algorithm. For each value of m and n, and each of n_1 in the range $(1, \min(m, n) - 1)$, s_1 in the range $(0, n_1)$, and r_1 in the range $(s_1 + 1, n_1)$, we determine the maximum of r in $(r_1 + 1, n)$ for each s in the range $(s_1 + 1, m)$, which satisfied the type II error constraints (5) and (6). We then check whether that set of parameters $(s_1, r_1, n_1, s, m, r, n)$ satisfied the type I error constraint (4). Taking the maximum r ensures not only the type I and type II error constraints being met (using the properties [b] and [c] of G being an increasing function of r and s), but also the type II error being as close as possible to the nominal level. We then compare among the feasible solutions to find the optimal solutions with respect to each of the four optimality criteria given previously. Finally we check the decreasing monotonicity of G in p for the optimal solutions to ensure that the type I and type II error constraints (1)–(3) are satisfied.

3. Numerical Results

For the utility of this article to practitioners we include two tables. In Tables 1 ($\alpha = 0.05$) and 2 ($\alpha = 0.10$), the response rates p_0 and p_2 are in the same range as that in Simon's paper. Of course, in Simon's design there was only one alternative p_* ($=p_1$ or p_2) allowed, while in our adaptive design p_* can be switched to p_1 from p_2 or vice versa, if necessary, after the first stage of the study. The true type I and type II errors attained by the four different optimality types and the expected sample sizes are presented. One can see that the type I error is generally controlled closely to the nominal level, and that in many cases the type II error, especially for β_2, is actually lower (hence higher power) than the nominal level due to the discreteness of the sample size.

Generally, as expected, the sample size of the first stage, n_1, is smaller and the total sample size m or n is larger for O1, as compared to the other three optimality types. The O3 and O4 designs may have similar configurations or even coincide, because they are selected within the more restrictive set where $\max(n, m)$ is smallest among all solutions. But they may also be very different, since O3 minimizes EN_0 while O4 minimizes $\max\{EN_i \mid i = 0, 1, 2\}$ in the restrictive set. For example, for $(p_0, p_1, p_2) = (0.30, 0.45, 0.50)$, O3 requires $r/n = 24/63$ but O4 requires only $r/n = 18/45$. The new O2 and O4 designs generally provide smaller expected sample size EN_1 and EN_2 (under the alternative hypothesis) than their counterparts O1 and O3 designs, respectively.

Note that, as reasoned previously, we set $\beta_1 \geq \beta_2$ (0.20 and 0.10, respectively) because we would like to have higher power for detecting more improvement of the new therapy (p_2 vs. p_0), and we need to lower the power for less improvement (p_1 vs. p_0) so that the sample size remains feasible for phase II cancer studies. For this reason and together with the discreteness of the sample size, the relative magnitude of m and n may not be consonant among optimality types and response rate cases. For example, in Table 1 for O1,

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Table 1

Designs for $\alpha = 0.05$, $\beta_1 = 0.20$, $\beta_2 = 0.10$, $p_1 - p_0 = 0.15$, *and* $p_2 - p_0 = 0.20$

			Design parameters			True			Expected sizes			Optimal
p_0	p_1	p_2	$s_1/r_1/n_1$	s/m	r/n	α	β_1	β_2	EN_0	EN_1	EN_2	type
0.05	0.20	0.25	0/2/9	3/31	5/43	0.049	0.200	0.094	17.23	31.19	34.14	1
			0/2/18	3/29	3/23	0.044	0.199	0.080	24.28	24.43	23.75	2
			0/1/21	2/26	3/26	0.047	0.197	0.076	24.30	25.95	25.99	3 & 4
0.10	0.25	0.30	2/3/18	6/38	8/49	0.048	0.199	0.077	24.40	42.93	45.99	1
			2/4/21	8/44	5/29	0.048	0.199	0.068	28.30	32.80	31.35	2
			1/3/18	6/37	7/38	0.050	0.199	0.068	28.54	36.94	37.57	3
			2/4/26	7/38	6/35	0.050	0.198	0.067	31.54	35.24	35.14	4
0.15	0.30	0.35	3/6/19	12/55	10/46	0.050	0.199	0.074	30.22	47.20	48.20	1
			4/7/27	12/51	8/34	0.049	0.198	0.061	35.48	39.57	37.27	2
			6/9/38	11/46	10/46	0.050	0.200	0.061	40.73	45.71	45.94	3 & 4
0.20	0.35	0.40	5/6/23	15/56	18/66	0.049	0.199	0.068	34.67	59.14	62.98	1
			8/11/35	17/60	12/40	0.049	0.196	0.058	40.69	45.81	43.25	2
			6/12/31	15/53	13/40	0.050	0.200	0.058	40.38	48.56	46.48	3 & 4
0.25	0.40	0.45	6/7/23	19/59	24/74	0.049	0.197	0.066	38.41	65.98	70.44	1
			9/14/38	22/65	15/41	0.050	0.200	0.058	50.32	50.18	45.62	2
			9/11/37	19/58	20/59	0.050	0.200	0.054	46.64	58.14	58.79	3
			9/16/42	20/59	17/51	0.050	0.200	0.058	52.53	54.58	52.81	4
0.30	0.45	0.50	9/10/28	22/58	29/77	0.049	0.200	0.061	41.16	69.38	73.94	1
			14/18/42	27/70	19/45	0.050	0.200	0.056	48.54	53.94	49.90	2
			12/16/39	25/64	24/63	0.050	0.199	0.051	48.50	62.11	62.87	3
			9/17/37	25/64	18/45	0.049	0.200	0.051	55.95	56.42	52.02	4
0.35	0.50	0.55	10/11/27	30/69	34/80	0.050	0.198	0.060	43.10	72.37	76.97	1
			17/20/43	38/87	21/45	0.050	0.200	0.057	50.66	56.09	50.70	2
			15/22/42	29/66	29/65	0.050	0.200	0.049	51.41	64.62	65.20	3
			19/26/50	29/66	29/62	0.050	0.200	0.049	54.36	63.71	63.36	4
0.40	0.55	0.60	11/12/26	38/79	39/82	0.050	0.200	0.062	43.89	74.13	78.93	1
			17/21/40	40/81	23/45	0.050	0.199	0.049	51.36	57.51	51.75	2
			14/19/36	34/69	32/66	0.050	0.200	0.047	51.77	66.11	66.43	3
			15/22/39	34/69	23/45	0.050	0.200	0.046	53.95	59.29	53.98	4
0.45	0.60	0.65	15/16/31	39/74	42/79	0.050	0.200	0.054	44.23	72.38	76.75	1
			20/25/43	41/75	26/46	0.049	0.199	0.044	53.68	57.68	51.92	2
			16/18/38	36/68	37/68	0.050	0.200	0.044	55.26	67.42	67.90	3
			19/27/46	37/68	35/66	0.050	0.200	0.044	59.97	66.79	66.44	4
0.50	0.65	0.70	16/17/30	40/69	46/79	0.049	0.200	0.052	43.21	71.88	76.59	1
			23/27/43	46/77	28/46	0.049	0.197	0.041	51.20	56.84	51.39	2
			26/32/53	39/66	40/67	0.050	0.200	0.042	59.55	66.56	66.90	3
			16/22/38	40/67	39/66	0.050	0.200	0.042	60.82	66.13	66.07	4
0.55	0.70	0.75	13/15/23	47/74	46/73	0.050	0.200	0.052	41.43	67.26	71.12	1
			22/26/38	48/74	27/41	0.050	0.200	0.039	47.83	53.98	47.79	2
			22/25/39	42/65	41/64	0.050	0.200	0.039	48.52	62.96	63.86	3
			22/30/43	42/65	31/48	0.050	0.200	0.039	56.78	57.09	52.46	4
0.60	0.75	0.80	13/16/21	49/72	46/67	0.050	0.199	0.051	38.64	63.54	66.87	1
			23/27/37	49/70	28/39	0.050	0.196	0.035	46.98	51.04	44.61	2
			22/25/37	43/61	42/61	0.050	0.198	0.035	48.15	60.35	60.92	3
			32/34/49	42/61	42/60	0.050	0.199	0.035	51.14	59.22	59.91	4
0.65	0.80	0.85	15/18/22	47/64	44/61	0.050	0.199	0.043	34.62	57.42	60.73	1
			23/26/33	47/63	27/36	0.050	0.198	0.032	39.13	46.29	41.28	2
			15/16/24	39/53	41/55	0.050	0.200	0.031	39.96	53.77	54.79	3
			22/28/34	41/55	30/38	0.050	0.199	0.032	43.32	49.33	44.79	4
0.70	0.85	0.90	14/16/19	46/59	41/52	0.049	0.199	0.039	29.97	50.13	52.65	1
			22/24/29	51/64	25/31	0.049	0.198	0.028	34.33	41.00	35.47	2
			13/14/19	37/47	39/49	0.049	0.199	0.026	32.83	47.21	48.69	3
			20/24/28	39/49	28/34	0.050	0.199	0.026	35.43	42.32	38.48	4
0.75	0.90	0.95	10/11/13	40/48	36/43	0.049	0.200	0.027	24.01	40.21	42.82	1
			19/21/24	42/49	22/26	0.049	0.200	0.012	29.25	33.89	28.52	2
			17/21/22	33/39	23/25	0.050	0.200	0.016	27.47	36.57	34.40	3 & 4

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Table 2

Designs for $\alpha = 0.10$, $\beta_1 = 0.20$, $\beta_2 = 0.10$, $p_1 - p_0 = 0.15$, *and* $p_2 - p_0 = 0.20$

			Design parameters			True			Expected sizes			Optimal
p_0	p_1	p_2	$s_1/r_1/n_1$	s/m	r/n	α	β_1	β_2	EN_0	EN_1	EN_2	type
0.05	0.20	0.25	0/1/9	2/23	2/27	0.092	0.199	0.100	14.46	23.38	24.75	1
			0/1/12	2/24	2/18	0.086	0.200	0.093	16.81	18.82	18.57	2
			0/1/17	1/18	2/20	0.091	0.200	0.087	18.00	19.74	19.89	3 & 4
0.10	0.25	0.30	1/2/14	4/26	5/34	0.095	0.199	0.085	20.25	30.54	32.14	1
			1/3/17	5/32	4/22	0.091	0.200	0.084	23.95	24.78	23.73	2
			1/3/20	5/28	4/27	0.099	0.200	0.080	24.73	27.03	27.05	3 & 4
0.15	0.30	0.35	2/3/15	7/34	9/45	0.098	0.199	0.085	24.47	39.32	41.93	1
			3/5/22	9/41	6/26	0.096	0.199	0.077	28.57	29.41	27.98	2
			2/5/26	8/34	7/33	0.100	0.200	0.075	31.98	33.11	33.05	3 & 4
0.20	0.35	0.40	3/5/17	10/39	14/52	0.099	0.200	0.076	28.30	44.28	47.55	1
			5/7/25	15/53	8/29	0.100	0.194	0.068	33.11	34.03	31.86	2
			4/5/23	10/40	11/40	0.098	0.200	0.070	31.49	39.06	39.68	3
			4/8/26	11/40	9/33	0.099	0.200	0.069	34.22	35.53	34.49	4
0.25	0.40	0.45	5/6/20	13/43	17/54	0.100	0.199	0.075	31.16	48.36	51.30	1
			7/10/28	16/49	11/32	0.097	0.199	0.068	35.25	37.22	35.12	2
			5/10/25	14/43	12/39	0.100	0.200	0.067	36.07	40.81	40.38	3
			4/10/24	14/43	11/33	0.099	0.200	0.067	38.10	39.25	37.47	4
0.30	0.45	0.50	6/10/20	20/55	16/44	0.100	0.199	0.075	33.53	47.71	48.45	1
			10/13/32	20/53	14/35	0.097	0.199	0.066	38.23	40.07	37.87	2
			8/11/29	17/46	18/47	0.099	0.200	0.064	37.99	45.99	46.66	3
			10/13/34	18/47	17/46	0.098	0.200	0.064	39.68	45.66	45.96	4
0.35	0.50	0.55	7/8/20	22/54	25/60	0.099	0.200	0.074	34.99	54.02	57.24	1
			11/14/30	25/58	15/34	0.097	0.199	0.063	38.10	41.46	38.61	2
			10/15/30	21/49	17/39	0.100	0.199	0.062	39.05	43.78	42.29	3
			9/15/29	21/49	16/34	0.099	0.200	0.062	40.54	43.05	40.31	4
0.40	0.55	0.60	9/13/22	27/58	28/60	0.100	0.200	0.070	35.57	53.77	56.92	1
			14/17/33	27/56	18/37	0.100	0.198	0.060	39.11	42.43	40.20	2
			9/14/25	24/50	23/50	0.100	0.200	0.060	39.38	48.90	49.67	3
			11/17/30	24/50	18/35	0.100	0.200	0.059	41.06	43.94	41.16	4
0.45	0.60	0.65	10/11/22	25/48	31/60	0.100	0.200	0.066	35.25	54.13	57.48	1
			15/18/32	33/62	19/34	0.097	0.197	0.060	40.35	42.33	38.67	2
			15/16/35	27/50	26/49	0.100	0.200	0.057	42.57	48.61	48.92	3 & 4
0.50	0.65	0.70	10/11/20	29/51	33/58	0.099	0.199	0.064	34.53	52.56	55.72	1
			15/19/30	31/53	20/34	0.099	0.200	0.052	38.90	41.85	38.73	2
			12/13/25	27/47	29/50	0.099	0.200	0.056	37.04	48.29	49.48	3
			19/23/36	29/50	24/39	0.099	0.199	0.055	39.97	43.34	41.59	4
0.55	0.70	0.75	11/12/20	26/42	36/58	0.098	0.198	0.059	33.15	51.86	55.47	1
			17/20/30	35/55	21/32	0.099	0.198	0.052	37.38	39.35	35.98	2
			13/18/27	29/46	30/47	0.100	0.200	0.052	40.39	46.31	46.74	3
			9/13/22	30/47	29/46	0.100	0.200	0.052	43.40	46.08	46.06	4
0.60	0.75	0.80	13/14/21	34/51	33/49	0.099	0.200	0.056	31.09	45.62	47.93	1
			17/20/28	36/52	21/30	0.098	0.198	0.047	35.94	37.18	33.64	2
			14/19/24	30/44	27/39	0.099	0.199	0.048	33.71	41.67	41.45	3
			17/22/28	30/44	23/31	0.100	0.200	0.049	34.23	39.48	37.26	4
0.65	0.80	0.85	12/15/18	34/47	18/26	0.100	0.199	0.052	27.80	37.45	35.71	1
			15/18/23	33/45	20/26	0.098	0.199	0.048	31.05	33.91	30.53	2
			12/16/19	30/41	20/27	0.100	0.198	0.044	29.35	36.20	34.46	3
			20/23/29	30/41	24/32	0.099	0.198	0.042	31.90	35.53	34.09	4
0.70	0.85	0.90	9/11/13	30/39	30/38	0.100	0.200	0.045	23.87	35.53	37.49	1
			15/17/21	36/45	18/23	0.100	0.200	0.033	27.82	29.57	26.00	2
			12/15/17	28/36	27/34	0.099	0.200	0.039	24.35	33.62	34.62	3
			17/20/23	28/36	23/28	0.100	0.198	0.037	26.37	31.99	30.97	4
0.75	0.90	0.95	8/10/11	24/29	26/31	0.100	0.197	0.025	19.28	28.02	29.86	1
			12/14/16	27/32	15/18	0.100	0.199	0.021	21.59	23.70	20.54	2
			13/15/17	24/29	24/28	0.099	0.198	0.023	21.19	27.53	28.10	3
			14/16/19	24/29	23/27	0.100	0.199	0.024	23.43	27.24	27.11	4

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m is smaller than n for $p \leq 0.05$ except for $(p_0, p_1, p_2) = (0.15, 0.30, 0.35)$. However, a closer examination of the algorithm reveals that the second-best solution for this case (not shown) has m smaller than n as for other cases, but with only a slightly larger EN_0 and higher β_2 than the best solution. This is because increasing either m or n will guarantee the required power. Therefore, sometimes we may want to examine the optimal solutions more closely, especially for O1 and O3 designs. In practice, our computer program (available at `http://www2.umdnj.edu/~linyo/adphii.htm`) provides users the option to list the 10 best feasible solutions for a given design configuration and the optimality type. This gives the user a better insight into the discrete searching process and more flexibility in choosing a design. The user may also choose a suboptimal solution, for whatever the reason, and still know the expected sample size and power at the two stages. Jung et al. (2001) also used a similar approach to find a good design that is a compromise between the optimal design and the minimax design for Simon's procedure.

Notice that the solutions to all the optimality types O1–O4 are unique, since the expected sample size EN_i is continuous. However, when we truncate EN_i to, say, the second decimal points for all practical purposes, as it was done in Tables 1 and 2, it is possible that it may appear that there are more than one optimal solution for O1 (minimize EN_0) and O2 (minimize max $\{EN_i;\ i = 0, 1, 2\}$). The chance is, however, quite small. We examined all the configurations in Tables 1 and 2, and found only one such case for each table. In Table 1, it was the case for O1 with the design: $p_0 = 0.55$, $p_1 = 0.70$, $p_2 = 0.75$, $\alpha = 0.05$, $\beta_1 = 0.20$, and $\beta_2 = 0.10$. In Table 2, it was the case for O2 with the design: $p_0 = 0.70$, $p_1 = 0.85$, $p_2 = 0.90$, $\alpha = 0.10$, $\beta_1 = 0.20$, and $\beta_2 = 0.10$. This phenomenon shows that the optimal and suboptimal solutions may sometimes differ very little in terms of the expected sample sizes. We recommend that the user utilize our computer program to examine in detail the top 10 feasible solutions and select one that fits best to the needs of the experimenter. In the following we continue to give a specific illustration of the adaptive design with the VBG example and compare it with Simon's design.

4. VBG Example Continued

For the VBG combination example, the uninteresting rate is 40% and the investigator's initial expectation of the VGB response rate is 60%. Hence, the testing of H_0: $p \leq 0.40$ is considered by traditional design with the type I error $\alpha \leq 5\%$ and type II error $\beta \leq 10\%$ at H_A: $p \geq 0.60$. Using Simon's two-stage minimax design, 29 patients will be recruited in the first stage of the study. The trial will be stopped with failure to reject null hypothesis H_0 if there are not more than 12 responses out of the 29 patients; otherwise, an additional 25 patients will be recruited in the second stage of the study. If there are not more than 27 responses out of the total of 54 patients, then there is not enough evidence to warrant further study (i.e., fail to reject null hypothesis).

However, suppose that the true response rate is actually 55% (i.e., 15% above p_0); the probability of not rejecting H_0 will be 29%, about three times as large as the chosen type II error. If the true response rate is 50% (i.e., 10% above p_0), the probability of not rejecting H_0 will be 57%, about six times the chosen type II error.

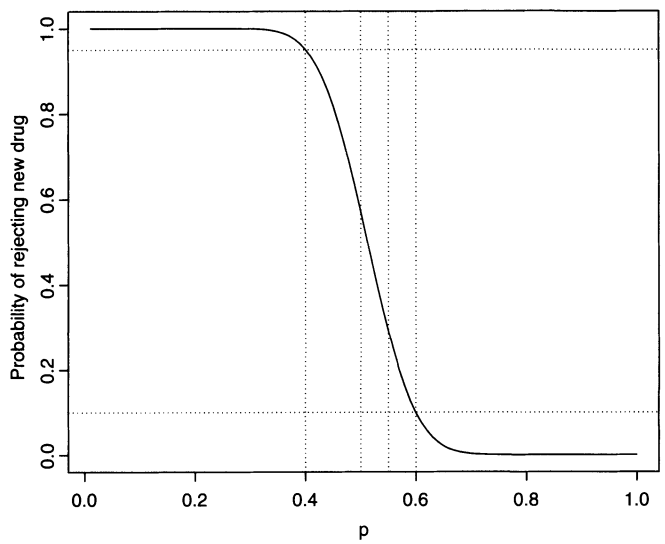

Figure 1. Probability of accepting H_0 when $p_0 = 0.40$, $p_* = 0.60$, $\alpha = 0.05$, and $\beta = 0.10$ from Simon's two-stage minimax design.

Using Simon's design, there is a very high probability to reject a promising new therapy if the initial expectation is set optimistically. This phenomenon can be demonstrated from Figure 1 for $\beta \leq 0.10$. In the figure, the curve between p_0 (=40%) and p_* (=60%) is very steep. When the true response rate decreases slightly from 60%, the probability of rejecting the therapy increases dramatically. A similar situation occurs when using Simon's optimal design. Using either Simon's optimal or minimax design, we are not able to switch to assuming H_A: $p \geq 0.55$.

If we consider testing of H_0: $p \leq 0.40$ with the type I error $\alpha \leq 5\%$ and type II error $\beta \leq 20\%$ at H_A: $p \geq 0.55$ from the very beginning with Simon's two-stage minimax design, 59 patients will be needed for the first stage of the study. The null hypothesis H_0 will not be rejected if there are not more than 28 responses out of the 59 patients; otherwise, an additional 11 patients are needed for the second stage of the study. The null hypothesis will not be rejected if there are not more than 34 responses out of the total of 70 patients.

Using the proposed adaptive two-stage design, the investigator is able to consider the two possibilities for response rate of the VBG combination, 55% (p_1) or 60% (p_2), as follows.

Suppose we choose $\alpha = 0.05$, $\beta_1 = 0.20$, and $\beta_2 = 0.10$. According to Table 1, for O4 (i.e., $\max\{EN_i \mid i = 0, 1, 2\}$ is smallest among such solutions that $\max(n, m)$ is smallest among all feasible solutions), we will enroll $n_1 = 39$ patients in the first stage. The number of patients to be recruited in stage II then depends on the number of responders out of this cohort of 39 patients. If 15 ($=s_1$) or fewer patients respond, the trial stops and we will conclude that there is not enough evidence for the response rate being greater than 40%. If 16–22 ($=r_1$) patients respond, then 30 additional patients will be recruited in the second stage. The total number of patients will be $m = 69$, and we are to test $p_0 = 0.40$ at 5% significance level versus

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$p_* = p_1 = 0.55$ with 80% power. If out of 69 patients the total number of responders is more than 34 ($=s$), then we reject the null hypothesis. If more than 22 patients respond out of the first cohort of 39 patients, then only 6 more patients will be needed for the second stage. The total number of patients will be $n = 45$, and we are to test $p_0 = 0.40$ at 5% significance level versus $p_* = p_2 = 0.60$ with 90% power. If out of the total of 45 patients there are more than 23 ($=r$) responders, then we reject the null hypothesis.

The above adaptive design and analysis will have a type I error rate of 5%. The attainable power will be 80% if $p \geq 55\%$, and at least 95% if $p \geq 60\%$. The expected number of patients (EN) is in the range of 54–60, depending on the true (unknown) probability of response, p. Of course, one can choose other optimality types O1–O3. The configuration will vary accordingly as shown in Table 1. For example, O1 design (i.e., minimizing EN_0) requires fewer patients at the first stage but more at the second stage.

Contrasting to Simon's two-stage design, the adaptive design requires 39 patients for the first stage, which is between 29 and 59 patients required for Simon's design for the two testing scenarios. The maximum number of patients in Simon's design is 54, compared to 45 with the adaptive design for H_A: $p \geq 0.60$, and is 70, compared to 69 with the adaptive design for H_A: $p \geq 0.55$. The attainable power is 90% for Simon's design and 95.4% for the adaptive design when testing for H_A: $p \geq 0.60$. More importantly, the adaptive design allows us to switch from H_A: $p \geq 0.60$ to H_A: $p \geq 0.55$ as indicated by the result of the first stage.

5. Discussion

Early phase cancer trials often assume the treatment effect with uncertainty. This article provides investigators a method to check the result at the first stage, adjust the power and response rate if necessary, and adapt the decision rule accordingly. Without this adaptability, a design such as Simon's would reject a potentially promising therapy if the expected response rate was too optimistic, or require more patients than necessary if the expected response rate was too conservative. The ability to detect a moderate gain in response rate, when the true response rate is lower than the initial expectation but still worthy, is an important feature of the adaptive design. To be able to save patient resources using the new optimality criteria (O2 and O4) is another useful contribution of this article. Simon's design is good for trials when the treatment effect specified in the alternative hypothesis is very certain. The adaptive design can be viewed as an extension of Simon's for situations when the treatment effect specified in the alternative hypothesis is not certain.

The two new proposed O2 and O4 designs generally require fewer patients than their Simon's counterparts for active therapies. With many optimality criteria available, our recommendation is to use our computer program to list the corresponding design parameters, sometimes even including suboptimal solutions, compare the figures, and select one that fits the study's resource and logistic constraints the best, and more importantly, note the statistical properties of the selected design.

Although some aspects of the proposed adaptive design are related to the idea of sample size reestimation (SSR) procedure, there is a clear distinction. In SSR, the response variance or the treatment effect to be detected after the first stage is usually the sample "reestimate" based on the interim data. In the proposed adaptive design for phase IIA cancer trials, p_* is not equal to the sample response rate at the first stage. The high (p_2) or low (p_1) value of p_* is predetermined. It is the selection between the high or low p_* that is adaptive and based on the result of the first stage. We do not extend the selection to more than two prefixed possible response rates mainly due to the complexity in the numerical solutions, and also because it is usually adequate for practitioners to contemplate between two (high/low) choices of p_*.

Finally, we note that we have used the hypothesis testing formulation in this article because this is the current framework employed by most authors in designing phase IIA investigations. As others have noted, estimation of the response rate may also be of interest when the drug is tested in later phase trials (Chen, 1997).

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Résumé

Un essai de phase IIa, sur une nouvelle thérapie anticancéreuse, consiste principalement à déterminer si cette thérapie s'avère suffisamment prometteuse dans le traitement d'un certain type de tumeur pour justifier la poursuite de son développement. On s'interdira, par la suite, toute nouvelle investigation, si le taux vrai de réponses est inférieur à un niveau jugé inintéressant, et on contrôle la puissance du test d'hypothèses pour un taux espéré, fixé a priori. Il est classique, dans une telle situation, d'utiliser des plans séquentiels en deux étapes. Cependant, au moment de l'élaboration du protocole, les investigateurs s'inquiètent souvent de l'incertitude entourant ce taux espéré, qui sert de base au calcul d'effectif. Dans cet article motivé par un exemple concret, nous proposons une stratégie adaptative de plans séquentiels en deux étapes. Cette stratégie—qui contrôle le risque de première espèce—utilise les résultats de l'analyse intermédiaire pour décider, soit le rejet de la nouvelle thérapie, soit la poursuite de l'essai sur la base d'une hypothèse optimiste ou pessimiste quant au taux de réponses espéré. Nous proposons enfin de nouveaux critères d'optimalité, quant à la réduction du nombre moyen de patients recrutés sur la totalité d'un essai.

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Appendix

A.1 *Proof of Properties of Function* $G(s_1, r_1, n_1, s, m, r, n, p)$

Proof of (a). First, we prove that for any k, t, and p, if $k/t \to q$ as $t \to \infty$, then as $t \to \infty$,

$$B(k,t,p) \to \begin{cases} 1 & \text{if } p < q, \\ 0 & \text{if } p > q. \end{cases} \tag{A.1}$$

Let $X_t \sim Bin(t, p)$, then by the Large Sample Theorem, $X_t/t \xrightarrow{P} p$ as $t \to \infty$. So, if $p < q$, for any sufficient large t such that $k/t > (p+q)/2$ and as $t \to \infty$,

$$\begin{aligned} B(k,t,p) &= P(X_t \le k) \\ &= P(X_t/t \le k/t) > P(X_t/t < (p+q)/2) \to 1. \end{aligned}$$

On the other hand, if $p > q$, then for any sufficient large t such that $k/t < p$ and as $t \to \infty$

$$B(k,t,p) = P(X_t \le k) = P(X_t/t \le k/t) < P(X_t/t < p) \to 0.$$

From (A.1), if $p < \min(q, q')$, then for any $s_1 + 1 \le x \le r_1$ and $r_1 + 1 \le y \le n_1$, since $(s-x)/m_2 \to q'$ and $(r-y)/n_2 \to q$ as $m \to \infty$ and $n \to \infty$,

$$B(s-x, m_2, p) \to 1, \quad \text{as } m \to \infty,$$

and

$$B(r-y, n_2, p) \to 1, \quad \text{as } n \to \infty.$$

Therefore, as $m \to \infty$ and $n \to \infty$,

$$\begin{aligned} G(s_1,r_1,n_1,s,m,r,n,p) &= B(s_1,n_1,p) \\ &\quad + \sum_{x=s_1+1}^{\min(r_1,s)} b(x,n_1,p)B(s-x,m_2,p) \\ &\quad + \sum_{y=r_1+1}^{\min(n_1,r)} b(y,n_1,p)B(r-y,n_2,p) \\ &\to B(s_1,n_1,p) + \sum_{x=s_1+1}^{r_1} b(x,n_1,p) \\ &\quad + \sum_{y=r_1+1}^{n_1} b(y,n_1,p) = 1. \end{aligned}$$

Similarly, if $q < p < q'$, then for any $s_1 + 1 \le x \le r_1$ and $r_1 + 1 \le y \le n_1$, since $(s-x)/m_2 \to q'$ and $(r-y)/n_2 \to q$ as $m \to \infty$ and $n \to \infty$,

$$B(s-x, m_2, p) \to 1, \qquad \text{as } m \to \infty,$$

and

$$B(r-y, n_2, p) \to 0, \qquad \text{as } n \to \infty.$$

So, as $m \to \infty$ and $n \to \infty$,

$$\begin{aligned} G(s_1,r_1,n_1,s,m,r,n,p) &= B(s_1,n_1,p) \\ &\quad + \sum_{x=s_1+1}^{\min(r_1,s)} b(x,n_1,p)B(s-x,m_2,p) \\ &\quad + \sum_{y=r_1+1}^{\min(r,n_1)} b(y,n_1,p)B(r-y,n_2,p) \cdot \\ &\to B(s_1,n_1,p) + \sum_{x=s_1+1}^{r_1} b(x,n_1,p) \\ &= B(r_1,n_1,p) \end{aligned}$$

The proofs are similar for the cases $q' < p < q$ and $p < \max(q, q')$.

Proof of (b). As s increases, $B(s-x, m_2, p)$ increases and $\sum_{x=s_1+1}^{\min(r_1,s)} b(x,n_1,p)B(s-x,m_2,p)$ increases, so does $G(s_1, r_1, n_1, s, m, r, n, p)$.

Proof of (c). As r increases, $B(r-x, n_2, p)$ increases and $\sum_{x=r_1+1}^{\min(r,n_1)} b(x,n_1,p)B(r-x,n_2,p)$ increases, so does $G(s_1, r_1, n_1, s, m, r, n, p)$.

A.2 *Proof of Existence of Solutions*

First, for any given p_1, p_2, β_1, and β_2, we can always find s_1 and n_1 such that

$$B(s_1,n_1,p_1) < \beta_1 \quad \text{and} \quad B(s_1,n_1,p_2) < \beta_2.$$

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Next, we choose q and q' such that $p_0 < q < p_1$ and $p_0 < q' < p_1$. Set $s = q'm$ and $r = qn$, then from property (b), when choosing m and n sufficiently large, the type I error constraint (1) will be satisfied because $G(s_1, r_1, n_1, s, m, r, n, p)$ will approach to 1 for $p \leq p_0 < \min(q, q')$ as m and n become large. Furthermore, the type II error constraints (2) and (3) will also be satisfied because $G(s_1, r_1, n_1, s, m, r, n, p)$ will approach to $B(s_1, n_1, p)$ for $p \geq p_1 > \max\{q, q'\}$ as m and n become sufficiently large. The above argument applies to any $r_1 > s_1$ as well.

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